

Manifold



M : line



M : plane

Slicing the thickness



Thick line in $\mathbb{R}^2$
Slice along the line

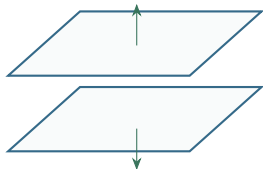


Board in $\mathbb{R}^3$
Slice along a plane

Orientation cover



$\widehat{M}$ has two components
Lines are orientable



$\widehat{M}$ has two components
Planes are orientable